Chemistry - Undergraduate Level Assessment 3 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following lists the hydrides of group-14 elements in order of thermal stability, from lowest to highest?
   (a) PbH 4 < SnH 4 < GeH 4 < SiH 4 < CH 4
   (b) PbH 4 < SnH 4 < CH 4 < GeH 4 < SiH 4
   (c) CH 4 < SiH 4 < GeH 4 < SnH 4 < PbH 4
   (d) CH 4 < PbH 4 < GeH 4 < SnH 4 < SiH 4

2. What is the limiting high-temperature molar heat capacity at constant volume (C_V) of a gas-phase diatomic molecule?
   (a) 1.5 R
   (b) 2 R
   (c) 2.5 R
   (d) 3.5 R

3. The T_2 of an NMR line is 15 ms. Ignoring other sources of linebroadening, calculate its linewidth.
   (a) 0.0471 Hz
   (b) 21.2 Hz
   (c) 42.4 Hz
   (d) 66.7 Hz

4. The normal modes of a carbon dioxide molecule that are infrared-active include which of the following? I. Bending II. Symmetric stretching III. Asymmetric stretching
   (a) I only
   (b) II only
   (c) III only
   (d) I and III only

5. Considering 0.1 M aqueous solutions of each of the following, which solution has the lowest pH?
   (a) Na_2CO_3
   (b) Na_3PO_4
   (c) Na_2S
   (d) NaCl

True / False (10 marks)

Answer True or False

6. True or False: A magnetic field of 9.49 T is optimal for the fastest spin-lattice relaxation for protons in a molecule with a rotational correlation time of 1 ns.

7. True or False: The nuclide 31 P has an NMR frequency of 115.5 MHz in a 20.0 T magnetic field.

8. True or False: HCl has the lowest melting point among the listed compounds due to its molecular nature and weak intermolecular forces.

9. True or False: Most organic compounds do not contain silicon atoms, making tetramethylsilane an ideal reference for 1 H NMR spectroscopy.

10. True or False: The molecular geometry of thionyl chloride, SOCl 2, is best described as trigonal pyramidal due to its lone pair of electrons.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that determine the strength of a 13 $C_{90}°pulseinnuclearmagneticresonance(NMR$

11. Discuss the concept of dissociation energy in the context of the hydrogen-bromine bond, detailing the enthalpy change associated with the reaction HBr(g) $\rightarrow$ H(g) + Br(g).

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